NeonPine – EMC design review & test plan

1. Standards (confirm the classification with the lab)

Market	Emissions	Immunity
EU / UK	**EN IEC 55015** (lighting equipment, decorative lighting is in scope)	**EN 61547** (lighting immunity)
EU (alternative classification)	EN 55032 Class B	EN 55035
USA	**FCC Part 15 Subpart B, Class B** (§15.107 conducted, §15.109 radiated, 30 MHz – 1 GHz)	–
Canada (option)	ICES-005 (lighting)	–

Test configuration: * the product in its final form (rev B with the rear cover) on a non-conductive table; * powered by a **representative CE/UL-listed 5 V USB-C adapter** through a 1 m USB-C cable. Conducted emissions are measured on the adapter's AC mains through a LISN; the adapter is support equipment. Record its model in the report and the manual; * firmware in **EMC mode**: all lamps lit, HV at 105 V, boost at its highest duty. Also scan a dimmed animation mode (75 V, BAM active) and standby.

2. Noise sources in the design

Source	Frequency content	Coupling path	Risk
Boost converter Q1/L11/D1	30 kHz fundamental, fast edges → harmonics up to ~100 MHz	**Magnetic field** from the boost loop (EN 55015 large-loop-antenna test, 9 kHz – 30 MHz); conducted on VBUS → USB cable → adapter	**High** – the 30 kHz fundamental and low harmonics sit inside the EN 55015 magnetic range
SW node (≈ 105 V swing)	edges ~20 ns	electric field from the SW copper area; USB cable as an antenna	Medium
MCU 48 MHz clock (HSI × 2)	48, 96, 144 … MHz	radiated via the USB cable and the board	Medium
BAM lamp switching	1 kHz frame, ~105 V edges through 150 k	low current → weak	Low
Touch pads (TKEY)	sampling	–	Low (emissions); **ESD entry point** (immunity)

3. Design measures already in rev A

- Shielded drum inductor (L11); boost input cap C3 next to L11/Q1, with a short return to the Q1 source (placement optimiser weights).
- Output loop D1 → C4 → GND kept tight (D1, C4, D2 grouped directly above the boost).

- GND pour on both layers; tented vias; decoupling C6 at U1 VDD; ADC filter C5 at U1.
- R1 47 Ω in the gate drive slows the turn-on edge; R2 keeps the boost off during reset.
- USB data lines not connected (power-only connector).

4. Additions for rev B

#	Measure	Cost	Helps	Rev B status
E1	**Ferrite bead (e.g. 600 Ω @ 100 MHz, 0603) in VBUS** after F1, plus the existing C1/C2 → a π filter	≈ $0.01	conducted emissions, radiated via the cable, 48 MHz harmonics	**done** (FB1)
E2	**Frequency dithering** of the 30 kHz boost (± 5 %, triangular, ~200 Hz) in firmware	free	spreads the boost harmonics (lowers quasi-peak)	firmware
E3	Optional footprint: RC snubber across Q1 D-S (e.g. 100 pF + 100 Ω)	≈ $0.01	SW-node ringing, 30–100 MHz	**footprint done** (R12/C8, DNP)
E4	GND stitching vias along the board edge and around the boost (every ~5 mm)	free	edge radiation	not done – add if the pre-scan needs it
E5	**Solder mask over TP1/TP2** (capacitive touch works through the mask)	free	ESD to the touch inputs	**done** (TP2 masked; TP1 removed)
E6	TVS on VBUS (e.g. 5 V uni-directional, SOD-323) at J1	≈ $0.02	ESD/surge on the connector	**done** (D3 SMF5.0A, 5.7 mm from VBUS)
E7	Keep the boost-off window as short as possible during standby wakeups	free	–	firmware

5. Immunity tests (EN 61547) and acceptance criteria

Phenomenon	Basic standard	Level (confirm with lab)	Where	Criterion
ESD	IEC 61000-4-2	± 4 kV contact / ± 8 kV air	touch pads, lamps, USB shell, edges	**B**: temporary change allowed, self-recovers; no HV above setpoint
Radiated RF	IEC 61000-4-3	3 V/m, 80 MHz – 1 GHz (+ bands to 6 GHz per edition), 80 % AM	whole unit	**A**: no visible flicker
Conducted RF (if cable > 3 m / adapter port)	IEC 61000-4-6	3 V	adapter AC port	A
EFT / surge / dips	IEC 61000-4-4/-5/-11	per standard (applied to the adapter's AC input)	via adapter	B / C

The lighting performance criterion is about light output: no visible change during the test (A), or a change that recovers by itself (B).

6. In-house pre-compliance (before booking the lab)

1. **Near-field scan** (H-field probe + spectrum analyser, 9 kHz – 1 GHz) over the boost area; compare 30 kHz harmonics with and without E2 dithering.
2. **Conducted**: an AC LISN in front of the adapter, 150 kHz – 30 MHz, compared with the Class B / EN 55015 limits minus a 6 dB margin.
3. **Radiated (rough)**: a biconical/log-periodic or calibrated near-field at 1 m in an office; look for 48 MHz harmonics and cable resonance; try clip-on ferrites to find the coupling path.
4. **ESD**: an ESD gun at ± 4/8 kV on the touch pads; check for resets, latch-up and HV overshoot on a scope (HV+ through a 100:1 HV probe).

7. Deliverables for the technical file

- Pre-compliance scan plots (rev B), the accredited lab report (EU + FCC), the test setup photo, adapter and cable identification, and the firmware version used.